Gowrish Kumar Reddy

Chennai, India

RESEARCH INTERESTS

Mechanistic Interpretability · AI Alignment & Safety · Efficient Edge AI Inference · Vision-Language Models

EDUCATION

SRM Institute of Science and Technology, Ramapuram *Chennai, India*
B.Tech, Computer Science & Engineering (AI & Machine Learning) CGPA 8.7 / 10.0 · Expected May 2028

RESEARCH PROJECTS

GPT-2 Small Mechanistic Interpretability Pipeline 2026

Python · PyTorch

- Built a complete GPT-2 Small forward pass from scratch – no interpretability libraries used.
- Engineered hook infrastructure for activation capture across all 144 attention head pairs.
- Injected synthetic weights to plant a Previous Token Head (L3H0) and Induction Head (L5H1); validated via activation patching (PTH score 0.87).
- Performed causal interventions via residual-stream patching to confirm circuit-level mechanistic behavior.

CPU-Only VQA Benchmark for Lightweight Vision-Language Models 2026

Python · PyTorch · HuggingFace Transformers · Matplotlib

- Benchmarked three sub-2B-parameter VLMs (SmolVLM-500M, Moondream2-1.8B, TinyLLaVA-1.4B) on a stratified 1,013-question VQAv2 subset under CPU-only, 8GB-RAM constraints, jointly reporting accuracy, latency, peak RAM, load time, and throughput - the first study to combine all five metrics for this model class on commodity laptop hardware.
- Discovered and statistically validated (chi-square test, $p < 0.000001$ across all three models) a consistent numeric-question accuracy collapse relative to yes/no and open-ended questions.
- Validated pairwise model comparisons via McNemar's exact test on paired predictions across 3,000+ CPU inference runs.
- Built a resumable, thermally-aware benchmark pipeline (automatic checkpoint/resume, scheduled cooldown breaks) enabling multi-day CPU-only evaluation on consumer laptop hardware.
- Manuscript in preparation for submission to a Scopus-indexed venue.

DistilBERT Sentiment Classifier 2026

Python · PyTorch · HuggingFace

- Fine-tuned DistilBERT-base on SST-2 with AdamW + cosine LR decay; achieved 89% test accuracy.
- Implemented the full training loop without the HuggingFace Trainer API for fine-grained optimization control.

Feedforward Neural Network from Scratch 2026

Python · NumPy

- Implemented backpropagation and gradient descent from first principles; solved XOR classification with zero external dependencies.

TECHNICAL SKILLS

Languages: Python, C
ML / DL: PyTorch, HuggingFace Transformers, scikit-learn
Research: Mechanistic Interpretability, Activation Patching, Causal Intervention, Attention Analysis
Tools: Git, GitHub, Linux, Jupyter, LaTeX, TransformerLens

RELEVANT COURSEWORK

Machine Learning · Deep Learning · Linear Algebra · Probability & Statistics · Data Structures & Algorithms · Computer Architecture · Operating Systems · Natural Language Processing